Aman Agrawal

Senior Research and Development Engineer

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Senior Research and Development Engineer specializing in large-scale generative and computer vision systems, with a focus on real-time 3D reconstruction, SLAM, and GPU-accelerated pipelines.

Building and scaling production-grade generative and vision systems at Flamapp.ai, deployed across global brands including Google, Samsung, Hyundai, YouTube, and Emirates Airlines, and used by a significant portion of Fortune 500 organizations.

Experienced in architecting and deploying end-to-end systems spanning model development, distributed inference, and cross-platform optimization. Strong background in systems-driven AI, with contributions across tracking, reconstruction, and real-time rendering pipelines.

Active in competitive engineering, with strong performance in robotics and systems competitions (Robocon 2020, top 16 virtual ranking), demonstrating consistent ownership of large-scale systems from concept to production.

Experience

Talking Avatar

Senior R&D Engineer · FLAMAPP.AI — *Jan 2026 – Present · Bengaluru*

- Architected and deployed end-to-end infrastructure for real-time talking avatar system using gRPC-based distributed inference.
- Fine-tuned and productionized models, improving latency, stability, and inference efficiency.
- Achieved <1.2s time-to-first-frame (TTFF) on dual RTX 4090 with scalable 1080p–4K output.
- Designed multi-model batching pipeline improving throughput and reducing generation cost to <$0.26/min.
- Contributed to system design integrated into production and company IP.

3D Reconstruction via Non-Markovian Process

Senior R&D Engineer · FLAMAPP.AI — *June 2025 – Aug 2025 · Bengaluru*

- Engineered core system for multi-view 3D reconstruction with high-fidelity textured mesh generation.
- Built differentiable rendering pipeline enabling faster and more stable camera parameter estimation.
- Implemented single-pass architecture for joint geometry and texture generation.
- Improved reconstruction fidelity using pretrained VAE with custom loss functions.
- Achieved 0.3px reprojection error and >95% text recovery accuracy.

Differentiable Keypoint Extraction & 6-DoF Tracking (Patent Filed)

R&D Engineer · FLAMAPP.AI — *July 2024 – Dec 2025 · Bengaluru*

- Co-developed production-grade SDK for real-time 6-DoF pose estimation (equal ownership).
- Built differentiable keypoint extraction pipeline outperforming ORB, SIFT, and BRISK.
- Implemented ICP-based refinement for robust and accurate pose estimation.
- Optimized system for edge deployment (<260KB), enabling production integration.
- Contributed to patented system deployed in a core company product.

Enterprise compliance platform

Full Stack Developer · Persistent Systems — *Dec 2021 – Sep 2024 · Pune*

- Modernized large-scale compliance platform improving system performance and reliability.
- Reduced error logs by ∼150k/week through architectural refactoring.
- Improved security and maintainability (SonarLint, Checkmarx).
- Replaced hardcoded configurations with environment-driven architecture.
- Resolved multiple Severity-1 production issues in enterprise systems.

Skills

Languages: C++23, Python, C#, JavaScript
Frameworks: OpenCV, WebGPU, Unity3D, .NET, ASP.NET MVC, Angular, Next.js
Systems: CUDA, GPU Computing, Docker, Linux/Unix, REST APIs
Domains: Computer Vision, Visual SLAM, 3D Reconstruction, NeRF, SIMD Optimization

Education

Shri Ramdeobaba College of Engineering and Management (RCOEM), Nagpur
B.E. Electronics Engineering — 2018 – 2022

Honors & Leadership

- Founding Engineer – Flying Flamingos – 2024-Present
- Member – .NET Foundation – 2021-2021
- President, Technical Club RCOEM – 2020-2022
- Member, Robotics Club RCOEM – 2019-2022
- 16th Place – Abu Robocon'2020 Virtual Ranking National Level – 2020
- Third Place – Autobot Robotics Competition – 2020

Personal Work

Algorithmic Trading System

C++ · Python · Quantitative Systems · Low-Latency Infra

- Designed and implemented end-to-end algorithmic trading pipeline with signal generation, backtesting, and execution layers.
- Developed dual-model conviction framework (bull/bear models) for high-confidence trade filtering.
- Built low-latency data processing and strategy evaluation pipelines for real-time decision making.

OpenPano – Real-time Panorama & 3D Stitching

Computer Vision · Geometry · Optimization

- Built panorama stitching system with feature matching, homography estimation, and image blending.
- Extended toward real-time and multi-view stitching with focus on geometric consistency and scalability.

Game Systems & Interactive Simulations

Unity3D · C# · Physics Systems

- Developed multiple game prototypes and systems focusing on gameplay mechanics, physics, and interaction design.
- Built simulation-driven environments for experimenting with real-time rendering and control systems.